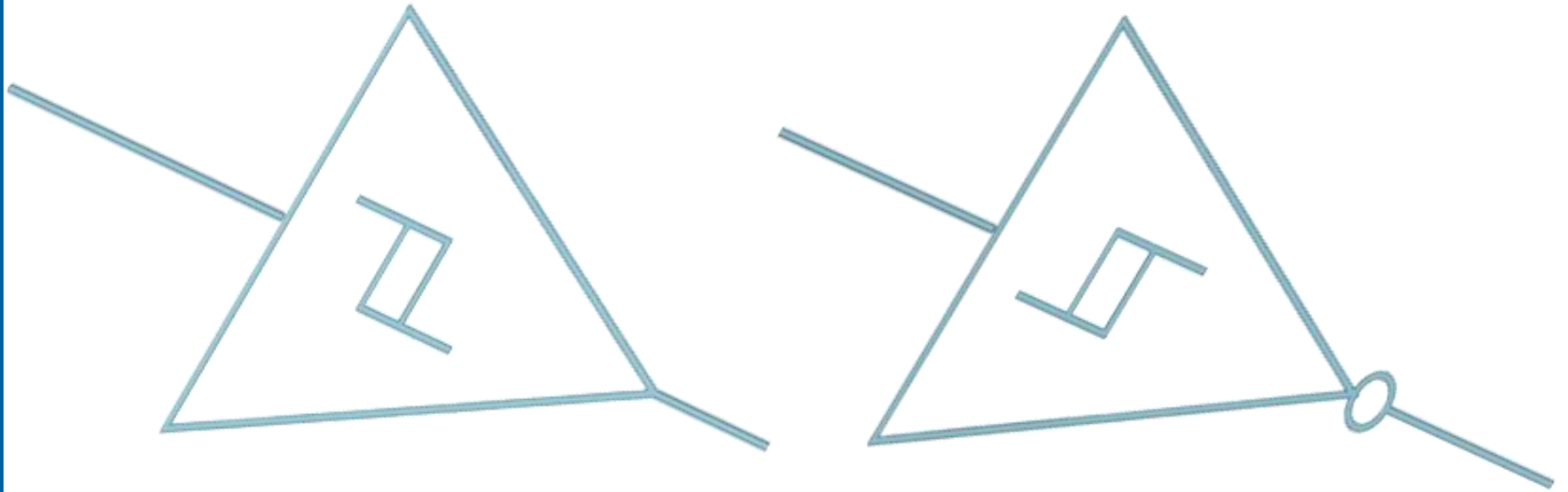




Non-linear Opamp Circuits

Comparators and Schmitt Triggers

Learning Objectives

- Understand the difference between linear and non-linear opamp circuits.
- Understand and analyse opamps used in the positive feedback configuration.
- Analyse and design Schmitt triggers and comparators.

Linear Circuits

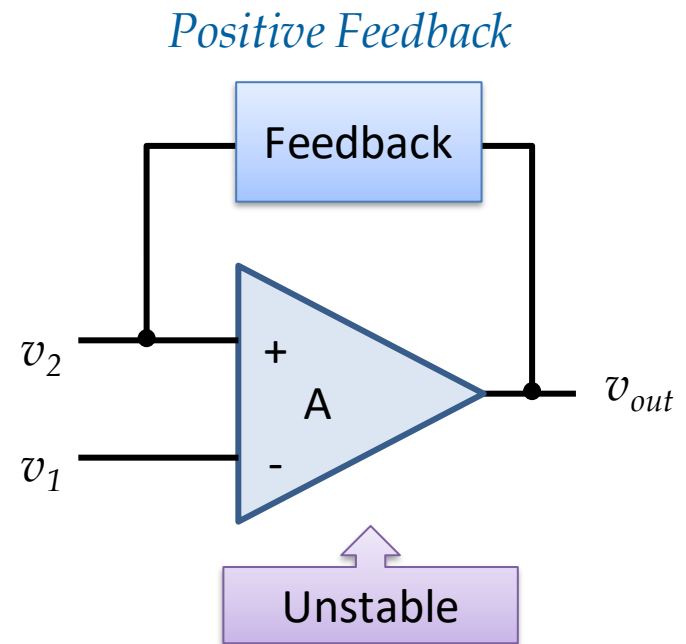
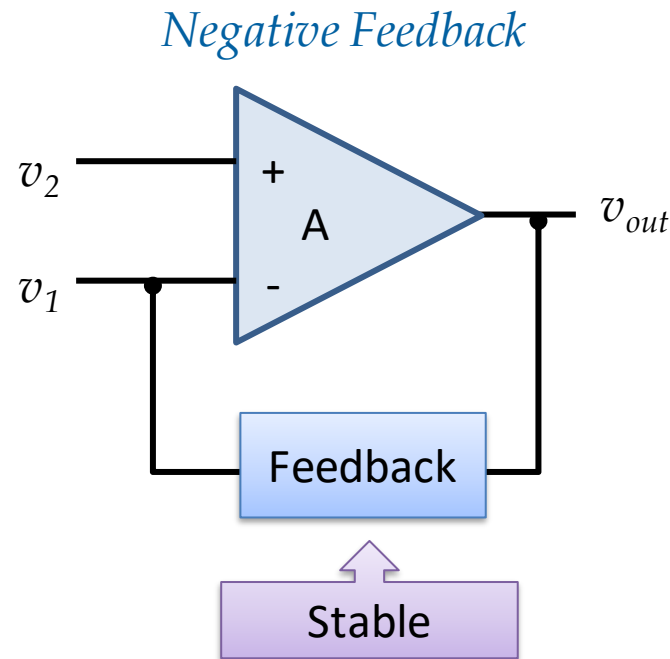
- Linear circuits produce an output which is a linear combination of its input signals.
- Linear circuit obey the superposition principle.
- They amplify and process electronic signals without distortion.
- The parameters of a linear circuit such as resistance, capacitance, inductance are constant and do not change with respect to the voltage and current.
- For example, if a sinusoidal input voltage or current of frequency f is applied at the input the steady-state output of the circuit is also sinusoidal with frequency f .
- Linear circuit examples: Amplifiers, differentiators, integrators, linear electronic filters etc.

Non-linear Circuits

- A nonlinear circuit's parameters such as resistance, inductance, capacitance, waveform and frequency are varied with respect to current and voltage and do not stay constant.
- Non-linear circuit can shape (or distort) an input signal, whether sinusoidal or not.
- Significant in both the frequency and time domains in circuits with saturation and feedback.
- For example, if a sinusoidal input voltage or current of frequency f is applied at the input the steady-state output cannot be expected to also be a sinusoidal with frequency f .
- Non-linear circuit examples: Schmitt triggers, comparators, modulators, rectifiers, radio receiver detectors, digital logic circuits.

Opamp Feedback Types

- Feedback loop between output and input can be connected either to the inverting or non-inverting terminal of the opamp.

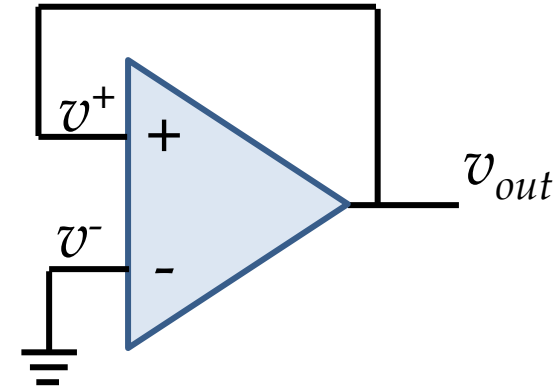


Negative Feedback

- Negative feedback allows to create all these practical circuits, with precisely set gains, rates, and other significant parameters.
- Negative feedback makes these circuits stable and self-correcting.
- The basic principle of negative feedback is that the output is driven to equilibrium (balance).
- With negative feedback, the output voltage “fed back” to the inverting input, the circuit tends to prevent itself from driving the output to full saturation and instead drives the output voltage only as high or as low as needed to balance the two inputs’ voltages.
- All opamp circuits we analysed so far have been in the negative feedback arrangement.

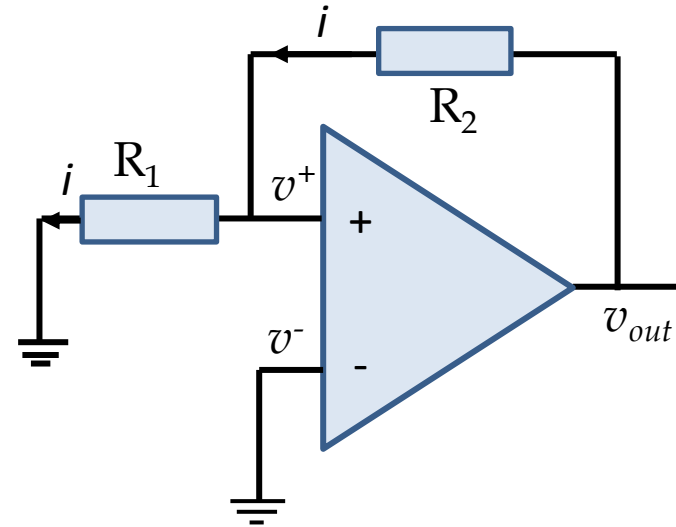
Simple Positive Feedback Circuit

- Connecting a straight piece of wire from output to the non-inverting input.
- Inverting input is disconnected from the feedback loop, and can be connected to any external voltage.
- If the non-inverting input voltage becomes positive, the op-amp will drive its output positive as well as the opamp internally makes $v_{out} = A (v^+ - v^-)$.
- The output voltage is fed back to the noninverting input in a back and forth iterative process, which will result in full positive output saturation.
- Similarly if the noninverting input starts out negative the op-amp's output will be driven to negative saturation.
- This is a *bistable* stable circuit with two possible states (saturated positive or saturated negative).
- Once the output reaches one of those saturated states, it will remain unchanged.



Bistable Circuit

- Electrical noise causes a small positive increment in the voltage v^+ .
- The large open loop gain will amplify this incremental signal and cause a much larger signal to appear at the opamp's output:
$$v_{out} = A (v^+ - v^-) = A v^+$$



- The voltage divider (R_1, R_2) will feed a fraction of v_{out} back to the non-inverting input terminal of the opamp where:

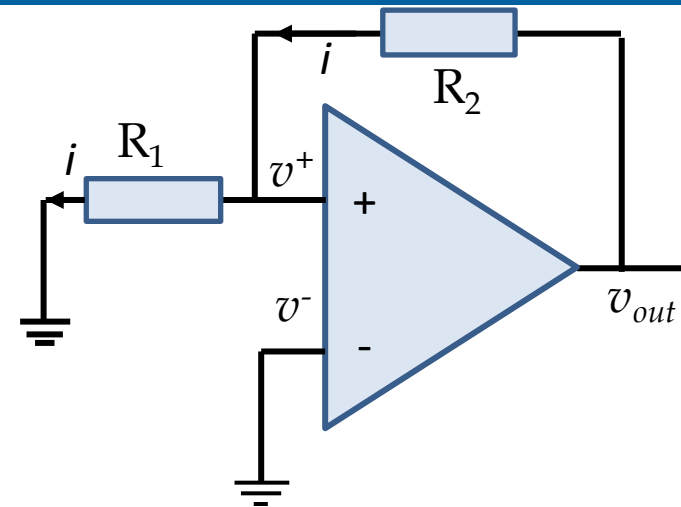
$$i = \frac{v_{out} - v^+}{R_2} = \frac{v^+ - 0}{R_1} \rightarrow v^+ = \frac{R_1}{(R_1 + R_2)} v_{out} = \beta v_{out}$$

Bistable Circuit

- The original incremental signal at v^+ is now amplified by $A\beta$ at the output, and if $A\beta$ is greater than unity (1) the fed-back signal will be greater than the original increment in v^+ .
- This regenerative process continues until eventually the opamp saturates with its output voltage at the positive-saturation level, L^+ , which is usually governed by the power supply rails of the opamp.
- When this happens, the voltage at the positive-input terminal, v^+ , becomes:

$$v^+ = \frac{R_1}{(R_1 + R_2)} L^+$$

Positive value – keeps opamp in positive saturation



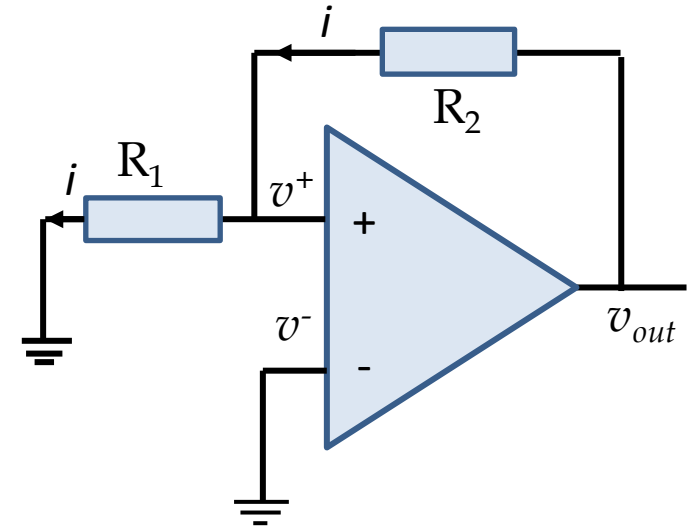
Bistable Circuit

- A near zero negative increment occurs at v^+ .
- The opamp output would saturate at negative saturation level L^- .
- When this happens, the voltage at the positive-input terminal, v^+ , becomes:

$$v^+ = \frac{R_1}{(R_1 + R_2)} L^-$$

Negative value – keeps opamp in negative saturation

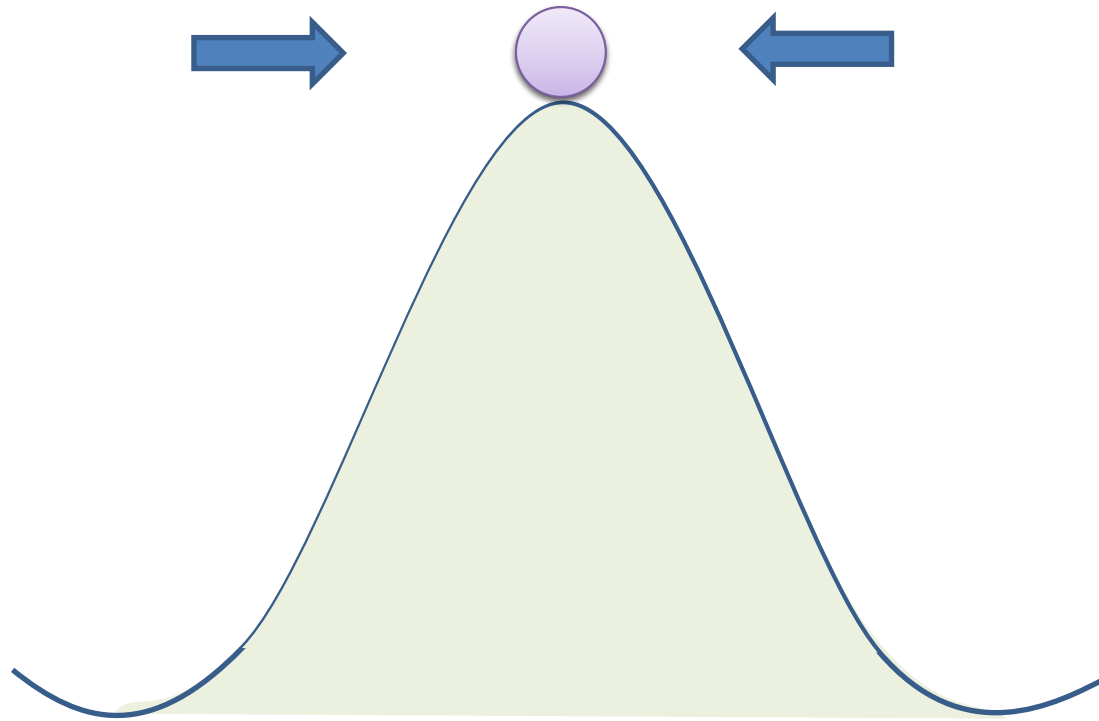
- This is the other stable state.



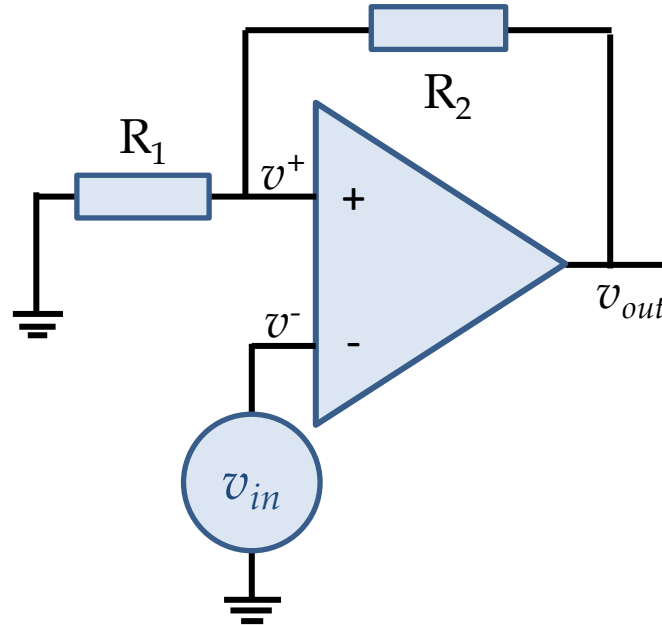
Bistable Circuit

- The circuit can exist in either of these two states indefinitely.
- Circuit cannot exist in the state for which $v^+ = 0$ and $v_{out} = 0$ for any length of time.
- This state of unstable equilibrium (also known as a metastable state) can be easily driven into one of the two stable states by any disturbance, such as electrical noise.
- In negative feedback circuits a virtual short circuit appears between the op amp's input terminals and maintains this virtual short circuit in the face of disturbances.

Bistable Circuit Analogy

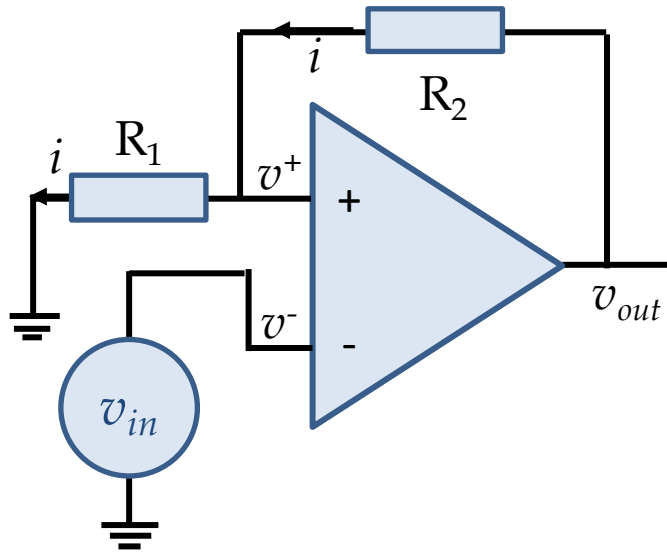


Inverting Schmitt Trigger



- Variations in the input voltage v_{in} causes the output alternate between two stable states.
- Increasing v_{in} causes the output to go negative and decreasing v_{in} causes the output to go positive.
- This Schmitt trigger displays inverting characteristics.

Inverting Schmitt Trigger



Due to infinite input impedance no current goes into the input terminals of op-amp:

$$i = \frac{v^+ - 0}{R_1} = \frac{v_{out} - v^+}{R_2}$$

$$v^+ = \frac{R_1}{(R_1 + R_2)} v_{out}$$

If circuit is positively stable at L^+ , negative values of v_{in} have no impact. To trigger circuit into L^- state, v_{in} must be made positive causing $(v^+ - v^-)$ to become negative.

This happens when $v^+ = v_{in} = V_{TH}$ & $v_{out} = L^+$

$$V_{TH} = \left(\frac{R_1}{R_1 + R_2} \right) L^+$$

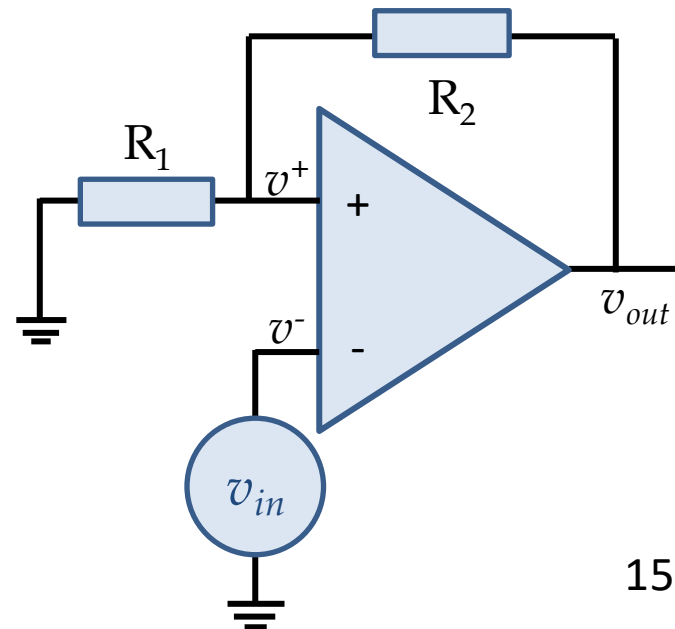
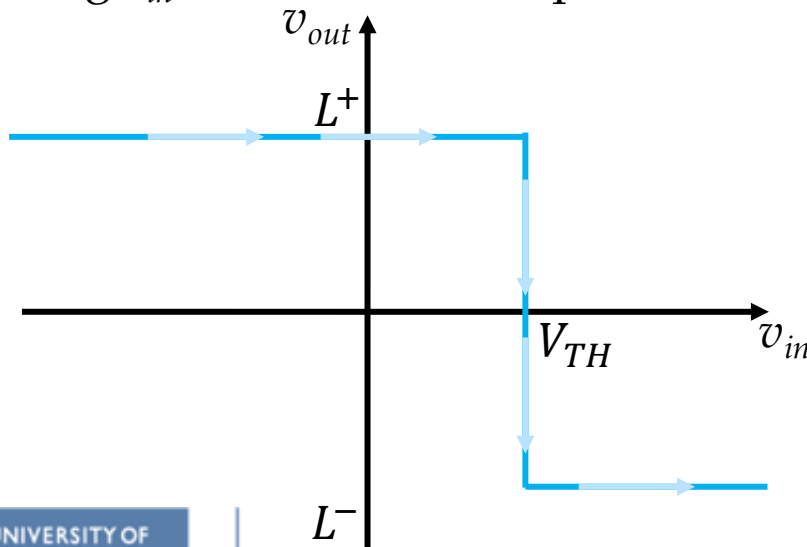
If circuit is negatively stable at L^- , positive values of v_{in} have no impact. To trigger circuit into L^+ state, v_{in} must be reduced causing $(v^+ - v^-)$ to become positive.

This happens when $v^+ = v_{in} = V_{TL}$ & $v_{out} = L^-$

$$V_{TL} = \left(\frac{R_1}{R_1 + R_2} \right) L^-$$

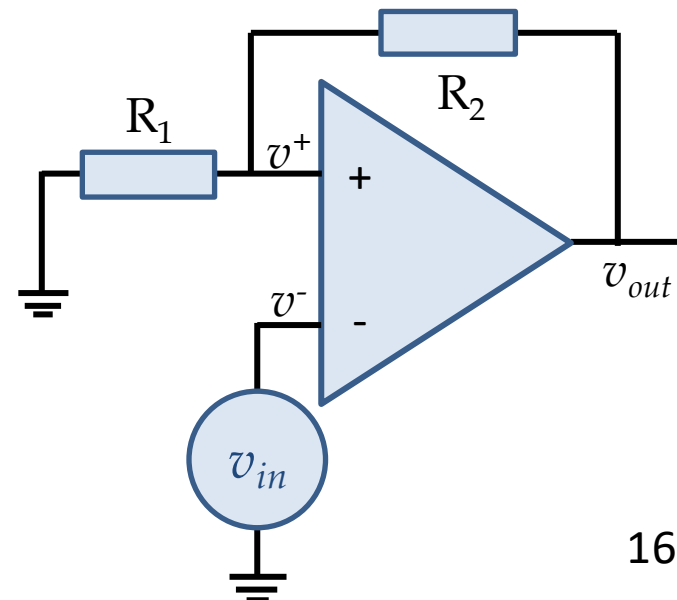
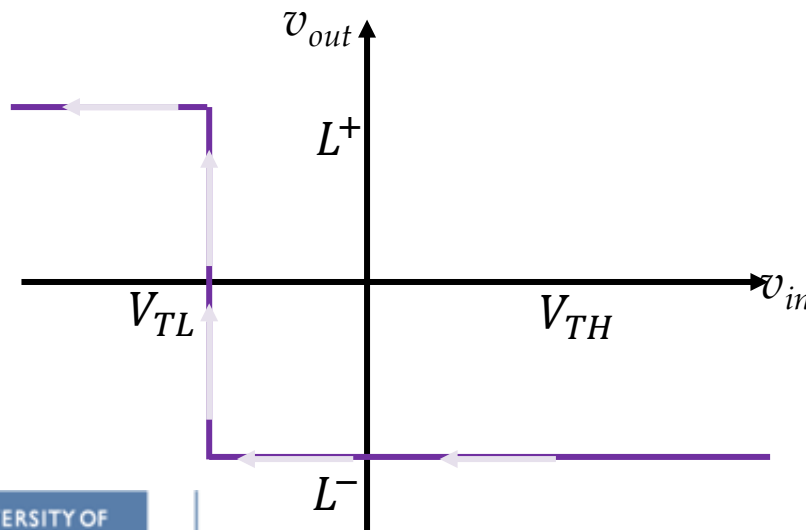
Transfer Characteristics

- Assume v_{out} is at one of its two possible levels, L^+ , and thus $v^+ = \beta L^+$.
- As v_{in} is gradually increased beyond $v^+ (= \beta L^+)$ the differential input to the opamp becomes negative and causes the output to become negative according to: $v_{out} = A (v^+ - v^-)$.
- The voltage divider then causes v^+ to go negative, thus increasing the net negative input to the opamp and keeps the regenerative process going.
- Opamp output saturates at L^- , and thus $v^+ = \beta L^-$.
- Increasing v_{in} further has no impact on the circuit's behaviour.



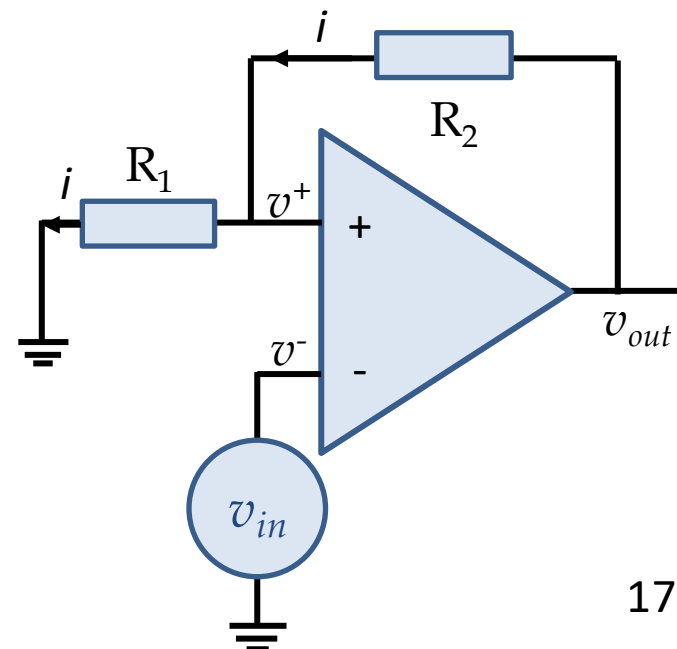
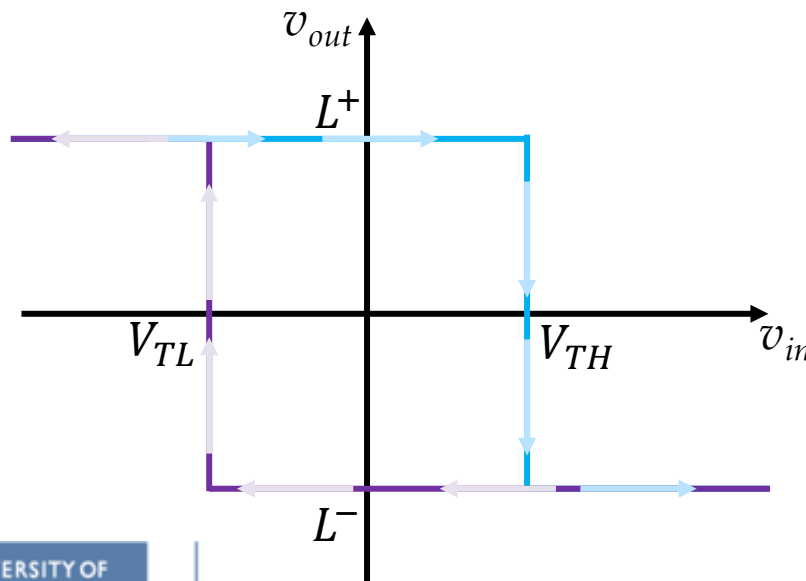
Transfer Characteristics

- v_{out} is now in its negative saturation level, L^- , and thus $v^+ = \beta L^-$.
- As v_{in} is gradually decreased beyond $v^+ (= \beta L^-)$ the differential input to the opamp becomes positive and causes the output to become positive according to: $v_{out} = A (v^+ - v^-)$.
- The voltage divider then causes v^+ to go positive, thus increasing the net positive input to the opamp and keeps the regenerative process going.
- Opamp output saturates at L^+ , and thus $v^+ = \beta L^+$.
- Decreasing v_{in} further has no impact on the circuit's behaviour.



Transfer Characteristics

- Complete transfer characteristics, $v_{out} - v_{in}$, of the circuit can be obtained by combining these two behaviours.
- Circuit changes state at different values of v_{in} depending on whether v_{in} is increasing or decreasing.
- Circuit is said to exhibit hysteresis and the width of the hysteresis is the difference between the high threshold V_{TH} and the low threshold V_{TL} .



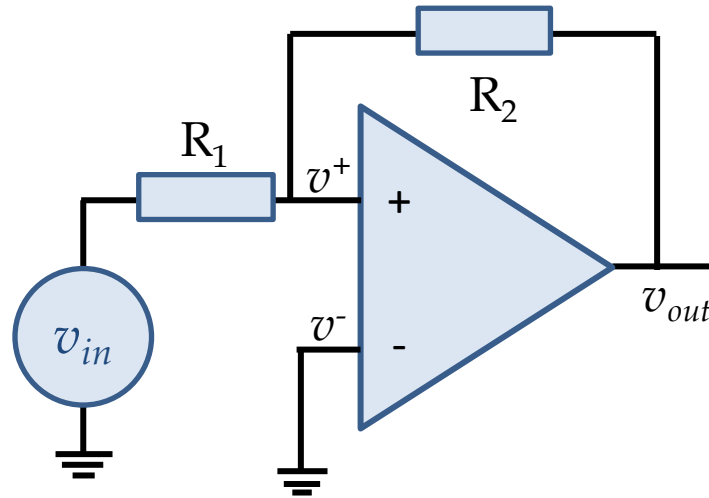
Triggering the Schmitt Trigger

- If the circuit is in the L^+ state it can be switched to the L^- state by applying an input v_{in} of value greater than the positive threshold voltage $V_{TH} = \beta L^+$.
- The net negative input between the opamp's input terminals triggers a regenerative cycle that drives the output to the stable state of L^- .
- Since v_{in} only triggers the switching process it can be a short pulse, or a trigger signal.
- Similarly applying a negative trigger signal at v_{in} of magnitude greater than the negative threshold voltage $V_{TL} = \beta L^-$.

Memory Element of Schmitt Trigger

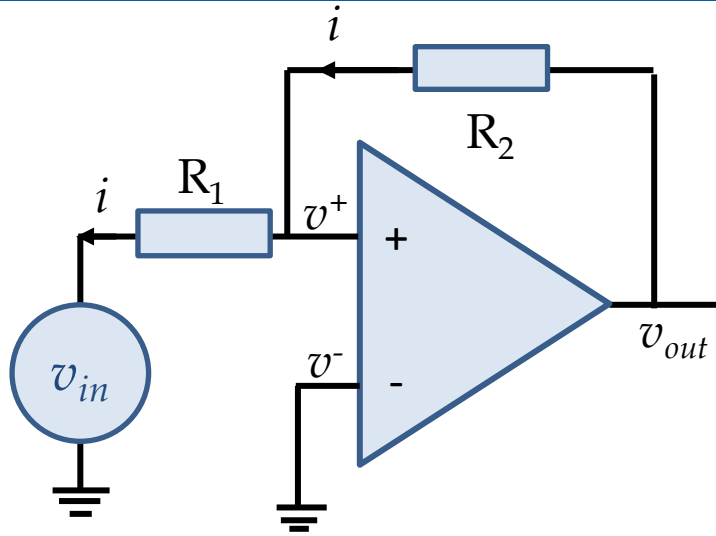
- For input voltages in the range $V_{TL} < v_{in} < V_{TH}$, the output can be either L+ or L-, depending on the state that the circuit is already in.
- The output is determined by the previous value of the trigger signal.
- The circuit exhibits memory.

Non-inverting Schmitt Trigger



- Variations in the input voltage v_{in} causes the output alternate between two stable states.
- Increasing v_{in} causes the output to go positive and decreasing v_{in} causes the output to go negative.
- This Schmitt trigger displays non-inverting characteristics.

Non-inverting Schmitt Trigger



Due to infinite input impedance no current goes into the input terminals of op-amp:

$$i = \frac{v^+ - v_{in}}{R_1} = \frac{v_{out} - v^+}{R_2}$$

$$v^+ = \frac{R_2}{(R_1 + R_2)} v_{in} + \frac{R_1}{(R_1 + R_2)} v_{out}$$

If circuit is positively stable at L^+ , positive values of v_{in} have no impact. To trigger circuit into L^- state, v_{in} must be made negative to cause v^+ to become negative.

This happens when $v^+ = 0$, $v_{out} = L^+$ & $v_{in} = V_{TL}$

$$V_{TL} = - \left(\frac{R_1}{R_2} \right) L^+$$

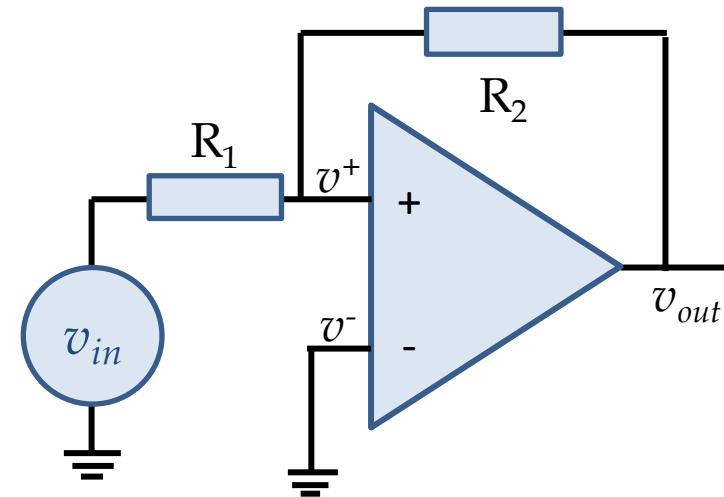
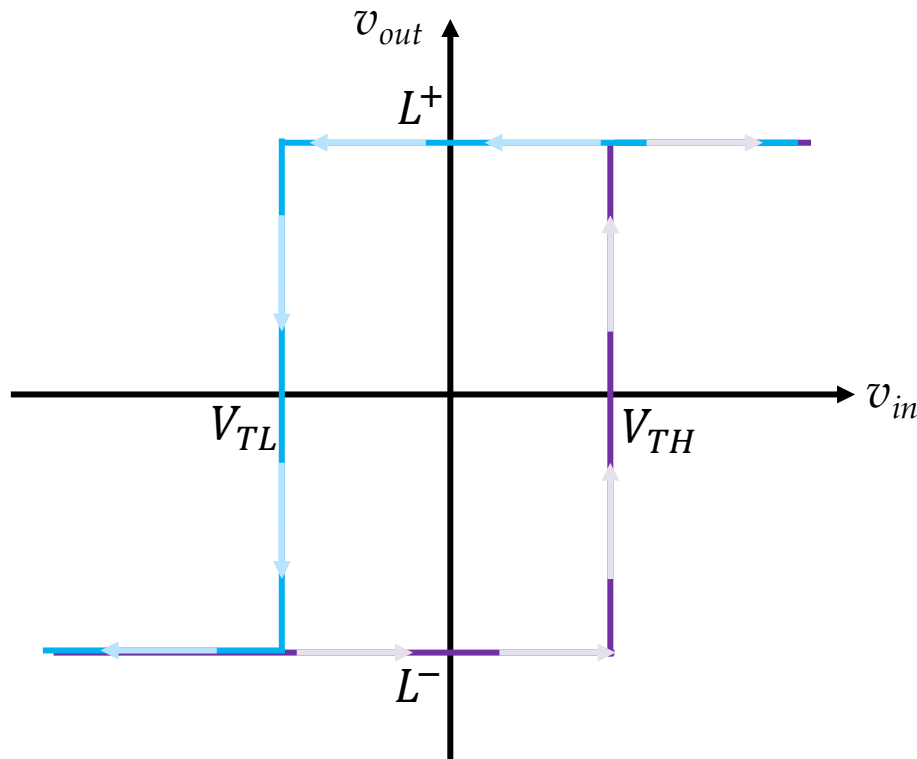
If circuit is negatively stable at L^- , negative values of v_{in} have no impact. To trigger circuit into L^+ state, v_{in} must be made positive to cause v^+ to become positive.

This happens when $v^+ = 0$, $v_{out} = L^-$ & $v_{in} = V_{TH}$

$$V_{TH} = - \left(\frac{R_1}{R_2} \right) L^-$$

Transfer Characteristics

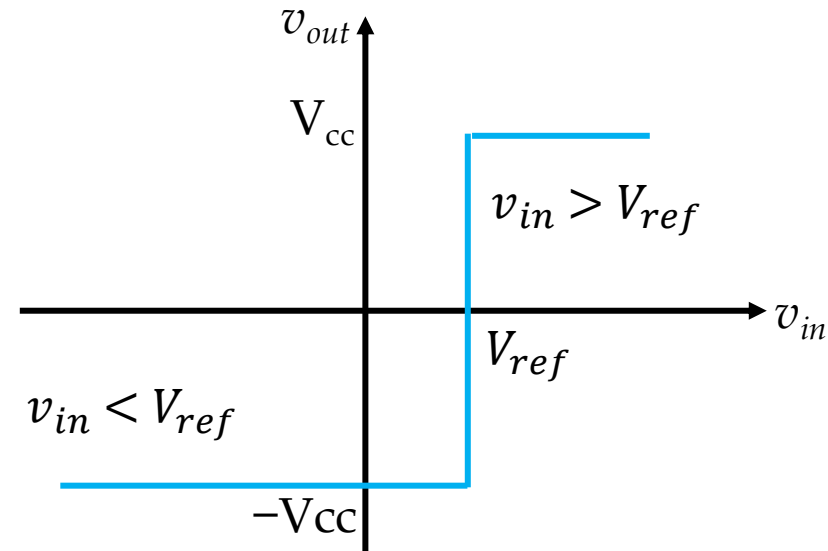
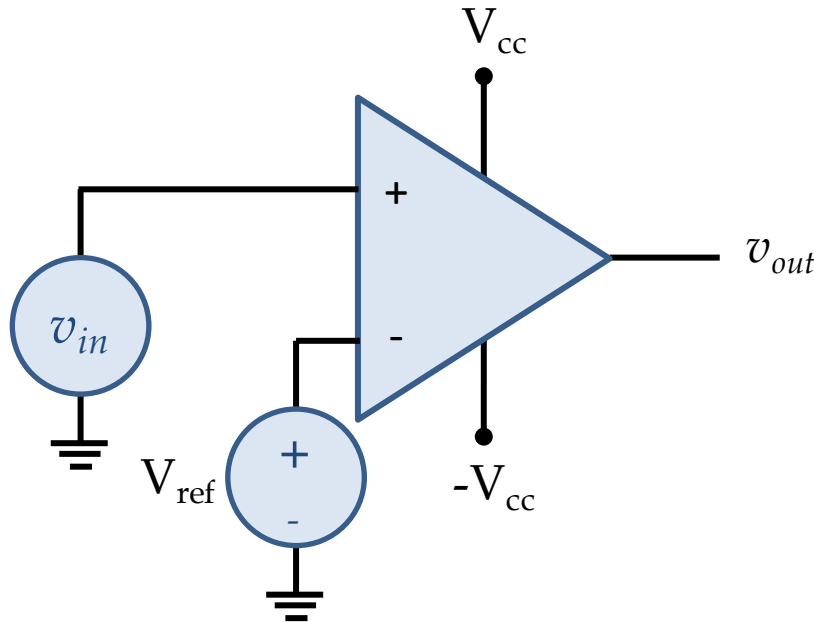
- Complete transfer characteristics, $v_{out} - v_{in}$, shows a non-inverting behaviour.



Comparator

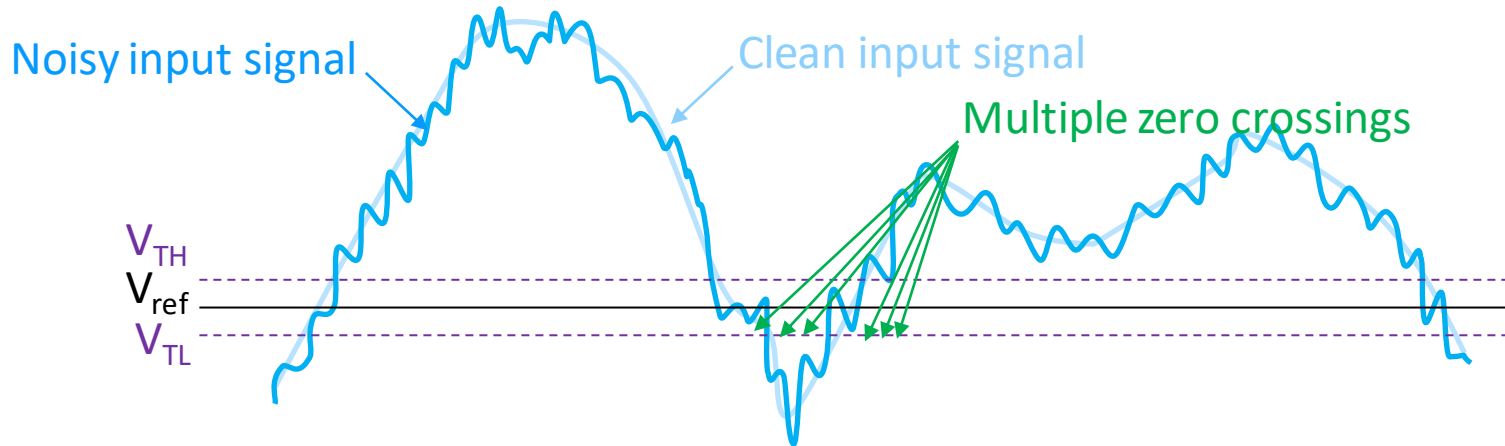
- A comparator compares one analogue voltage level with another, or a preset reference voltage, V_{REF} and produces an output signal based on this voltage comparison.
- Compares the magnitudes of two voltage inputs to determine which is the largest of the two.
- Comparators are used to detect the level of an input signal relative to a preset threshold value, to design of analog-to-digital (A/D) converters, detect zero crossings in a signal etc.

Comparator – Open Loop Configuration



- Suitable for applications where input signal varies rapidly or is not too noisy.
- For slow varying inputs or (electrically) noisy environments it may oscillate when the input is in the range of the reference voltage.

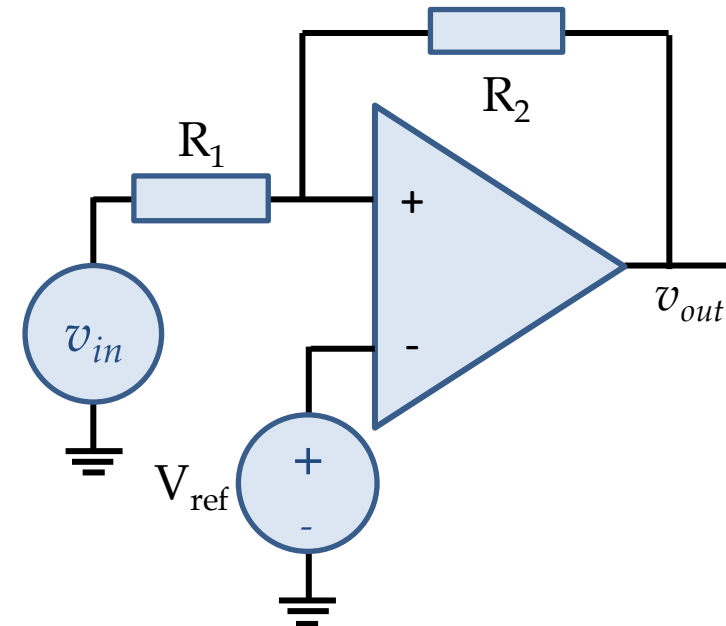
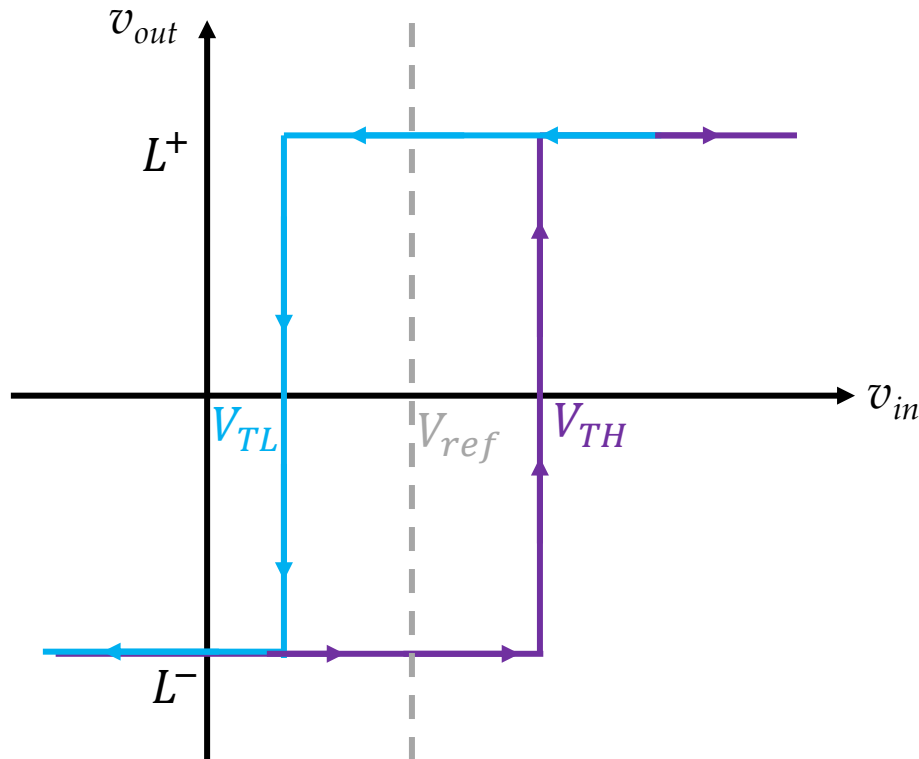
Comparator



- If V_{ref} was set to 0, the same circuit should detect zero crossings in the signal.
- Signal might cross the zero axis a number of times around each of the zero-crossing points we are trying to detect due to noise.
- If the expected peak-to-peak amplitude of the interference is known introducing hysteresis of appropriate width in the comparator characteristics will resolve the issue.

Comparator with Feedback

- V_{ref} , V_{TH} and V_{TL} need to be designed according to the requirements of the comparator.



Comparator with Feedback

- If the input signal is increasing in magnitude the comparator with hysteresis will remain in the low state until the input level exceeds the high threshold V_{TH} .
- Subsequently the comparator will remain in the high state even if, owing to interference, the signal decreases below V_{TH} .
- The comparator will switch to the low state only if the input signal is decreased below the low threshold V_{TL} .